

Influence of Inertia and Viscous Damping on the Magnetic Hysteresis and First-Order Reversal Curve (FORC) Diagrams of Compass-Needle Lattices

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The collective behavior of two-dimensional arrays of magnetic dipoles has attracted sustained interest in the context of artificial spin ice (ASI) and nanostructured magnetic islands fabricated by nanolithography. Classic thermodynamic studies of compass-needle lattices [1] and the introduction of First-Order Reversal Curve (FORC) diagrams [2] established the conceptual tools now widely used to map coercive and interaction field distributions in magnetically ordered arrays. However, existing models rely on quasi-static or Monte Carlo approaches that omit physical rotational inertia (I) and viscous damping (b), effectively assuming an infinite-damping limit ($Q \rightarrow 0$); variable FORC sweep rates further introduce relaxation artifacts that distort coercive-field scaling and mask cooperative switching cascades [3]. We address these gaps with a Newtonian rotational molecular-dynamics model—integrated by a Velocity-Verlet scheme—in which each needle obeys $I\ddot{\theta}_i = \tau_i^{\text{dip}} + \tau_i^{\text{ext}} - b\dot{\theta}_i$, with I auto-calibrated from a rectangular blade plus a central cylindrical brass pivot. Hysteresis loops and FORC diagrams are computed for square, triangular, and honeycomb geometries across a wide range of quality factors $Q = \omega_0 I/b$, and under a strictly constant field sweep rate $R = |dB/dt|$ that eliminates rate-artifact contamination. Inertia delays switching, dynamically inflates the coercive field, and generates phase-coherent oscillations at high Q ; the resulting FORC broadening mimics structural disorder. Crucially, the mechanical damping ratio b/I maps directly onto the phenomenological Gilbert damping parameter α of thin-film nanomagnets, establishing this macroscopic system as a transparent analog for nanolithography-fabricated magnetic island arrays and providing quantitative guidance for interpreting FORC measurements on nanopatterned samples.

References

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